25901129 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOTA08 - ARTIFICIAL INTELLIGENCE IN PLANT SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Mention any two types of data used in AI. [K] / [CO1]
2. How does CNN detect plant diseases? [U] / [CO2]
3. What is climate resilience in agriculture? [K] / [CO3]
4. Define transparency in AI systems. [K] / [CO2]
5. Why is smart irrigation essential in precision agriculture? [U] / [CO3]
6. Name one of the earliest AI programs. [K] / [CO1]
7. How will you collect images for disease dataset? [An] / [CO3]
8. Name one technology used to collect field data. [K] / [CO2]
9. Assess one advantage of AI used by startups. [E] / [CO3]
10. What is accountability in AI? [K] / [CO4]

11. How does a plant ID app identify plants? [U] / [CO2]

12. Why is preprocessing done in image analysis? [U] / [CO2]

[2x10 = 20]

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. Evaluate limitations of AI agricultural platforms. [E] / [CO3]

14. Explain why data privacy is included in ICAR guidelines. [U] / [CO2]

15. Explain the workflow of AI plant disease detection. [U] / [CO3]

16. Write short notes on the role of Artificial Intelligence in agriculture. [K] / [CO1]

17. What is plant stress? [K] / [CO1]

18. Explain how AI predicts crop yield. [U] / [CO2]

19. Suggest data governance measures for rural digital farming platforms. [A] / [CO4]

20. Define unsupervised learning. [K] / [CO1]

21. Define OpenCV and its importance. [K] / [CO1]

22. Suggest steps to get correct plant disease output from an app. [A] / [CO4]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Explain how AI supports nutrient prediction and recommendations. [U] / [CO3]

24. Explain why IEEE initiatives are important for global AI progress. [U] / [CO2]

[10x1 = 10]